

Junbo (Steven) Huang

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EXPERIENCE

Ocado Technology

Toronto, ON

Software Engineer II - Product & Applied ML

Jun 2024 - Present

- Led development of ML-based perception software for a 6-DoF robotic manipulation system, directly enabling robust, high-throughput warehouse automation and fulfillment at scale.
- Engineered end-to-end ML pipelines spanning data collection, labelling, model training, and metrics logging across BigQuery, Labelbox, Vertex AI, and ClearML, reducing time-to-production for new model iterations.
- Trained and fine-tuned Mask2Former in PyTorch, achieving 20%+ improvement over baseline AP/AR metrics through iterative dataset curation and hyperparameter optimization; exported and deployed versioned models for production serving.
- Implemented a centralized inference server abstracting Nvidia Triton and TorchServe backends, optimizing GPU resource allocation and inference scheduling to reduce system memory usage by 30%+.
- Led multiple epics from discovery to delivery within tight timelines by aligning stakeholder expectations and engineering constraints, regularly contributing to system architecture design, code reviews, and CI/CD.
- Designed and implemented robotic grasping and placement algorithms in C++ and Python using OOP principles, increasing item handling throughput (Units Per Hour) by 70%+ with a 92% autonomous pick rate in production.
- Architected 5+ gRPC microservices, deployed to production using Docker/Nomad while maintaining high system availability.

Ocado Technology

Toronto, ON

Machine Learning Engineer Intern

May 2023 - Dec 2023

- Researched and evaluated SOTA deep learning models for instance segmentation and video object tracking to improve accuracy and throughput of a high-speed robotic picking system.
- Prototyped an item trajectory tracking system achieving 96% prediction accuracy by combining SOTA tracking methods with novel visual feature embedding similarity scoring, reducing redundant robot pose computation by 65%.
- Optimized model inference speed by 50% through quantization, pruning, and graph optimization using ONNX and TensorRT.

McMaster University

Hamilton, ON

Research Engineer Intern

May 2022 - Sept 2022

- Developed an edge AI PoC for traffic monitoring using Nvidia DeepStream on the Jetson platform, building end-to-end pipelines covering camera ingestion, inference, and real-time analytics.
- Fine-tuned YOLOv4 via transfer learning for real-time vehicle and pedestrian object detection, achieving 93% accuracy at 30+ FPS in edge-constrained compute environments.

IBM

Markham, ON

Software Developer Intern

May 2020 - Aug 2021

- Developed compiler modules in C/Java and implemented profiling tools in Python to automate performance analysis across 20+ release versions, surfacing regression trends for engineering teams.

EDUCATION

University of Toronto

Sept 2022 - Jan 2024

M.Sc in Applied Computing, Computer Science

(GPA: 3.94/4.0)

- AI Concentration (Deep Learning, Computer Vision, Computing Systems, Cloud Data Analytics)

McMaster University

Sept 2017 - Jun 2022

B.Eng, Mechatronics Engineering

(GPA: 3.97/4.0)

SKILLS

Languages: Python, C++, Golang, SQL, Java, Bash

Frameworks: Docker, Google Cloud Platform, Nvidia Triton Server, Nvidia DeepStream, TorchServe, Ollama

Tools: Git/GitLab, BigQuery, Vertex AI, Nomad, ClearML, Pytest

ML: PyTorch, JAX, HuggingFace, ONNX, TensorRT, Scikit-Learn, NumPy, OpenCV, Transformers, PEFT

PROJECTS

LLM-powered Texas Hold'em Game

- Architected a multi-threaded C++17 Texas Hold'em engine with a 4-layer architecture (UI, Players, AI, Core), enforcing separation of concerns via abstract interfaces and thread-safe state management.
- Engineered LLM-powered AI opponents via an OpenAI-compatible REST API, enabling pluggable local inference backends (Ollama, llama.cpp) and personality-driven structured generation through dynamic system prompt injection.

CNN Adversarial Defense

- Implemented a U-Net-based Encoder-Decoder image denoising network in PyTorch to denoise adversarially perturbed images.
- Applied model ensemble techniques to improve CNN classifier robustness against adversarial attacks, boosting ResNet classification accuracy by 21% on adversarially targeted images.